ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 04

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The point (−6, −2) lies in the:

(a) fourth quadrant (b) first quadrant

(c) third quadrant (d) second quadrant

Ans: (c) Both coordinates are negative ⇒ third quadrant.

2. The point on the y-axis which is 5 units below the origin is:

(a) (0, −5) (b) (5, 0) (c) (0, 5) (d) (−5, 0)

Ans: (a) On the y-axis x = 0; below the origin means y is negative.

3. The solution of the pair of equations x + y = 5 and x − y = 1 is:

(a) x = 2, y = 3 (b) x = 4, y = 1 (c) x = 3, y = 2 (d) x = 5, y = 0

Ans: (c) Adding: 2x = 6 ⇒ x = 3; then y = 2.

4. Which of the following is an irrational number?

(a) 0.1010010001… (the number of zeros keeps increasing)

(b) 0.25

(c) 3.14

(d) 0.101010… (10 repeating)

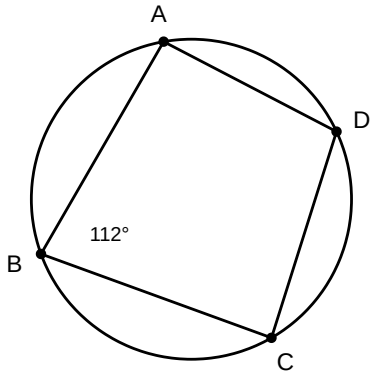
Ans: (a) A non-terminating, non-repeating decimal is irrational.

5. The product $(x - 3)(x^2 + 3x + 9)$ is equal to:

(a) $x^3 - 9$ (b) $x^3 + 27$ (c) $(x - 3)^3$ (d) $x^3 - 27$

Ans: (d) Using $(a - b)(a^2 + ab + b^2) = a^3 - b^3$ with a = x, b = 3.

6. ABCD is a cyclic quadrilateral with ∠ABC = 112°. Then ∠ADC is:



(a) 112° (b) 98° (c) 78° (d) 68°

Ans: (d) Opposite angles of a cyclic quadrilateral are supplementary: 180° − 112° = 68°.

7. The perpendicular from the centre of a circle to a chord is 12 cm long and the chord is 18 cm long. The radius of the circle is:

(a) 30 cm (b) √63 cm (c) 15 cm (d) 21 cm

Ans: (c) Half chord = 9 cm; $r^2 = 12^2 + 9^2 = 225 \Rightarrow r = 15$ cm.

8. An isosceles triangle has equal sides of 17 cm and base 16 cm. Its area, by Heron's formula, is:

(a) 120 cm² (b) 136 cm² (c) 240 cm² (d) 60 cm²

Ans: (a) $s = 25$; area $= \sqrt{(25 \times 8 \times 8 \times 9)} = 120$ cm².

9. A card is drawn at random from cards numbered 1 to 20. The probability that it is a prime number is:

(a) $\frac{2}{5}$ (b) $\frac{9}{20}$ (c) $\frac{1}{2}$ (d) $\frac{3}{10}$

Ans: (a) Primes ≤ 20: 2, 3, 5, 7, 11, 13, 17, 19 = 8 ⇒ 8/20 = 2/5.

10. If $t_n = (-1)^n(3n - 1)$, then t_5 is:

(a) 14 (b) −14 (c) 16 (d) −16

Ans: (b) $t_5 = (-1)^5 \times 14 = -14$.

11. The converse of 'If two lines are parallel, then the corresponding angles formed by a transversal are equal' is:

(a) If the corresponding angles are not equal, the lines are parallel.
(b) If the corresponding angles formed by a transversal are equal, then the two lines are parallel.
(c) If two lines are not parallel, then the corresponding angles are not equal.
(d) Parallel lines have equal alternate angles.

Ans: (b) Swap the 'if' part and the 'then' part.

12. A counterexample to 'If a number ends in 5, then it is a multiple of 10' is:

(a) 10 (b) 30 (c) 20 (d) 15

Ans: (d) 15 ends in 5 but is not a multiple of 10.

13. A class has 12 boys of average height 150 cm and 18 girls of average height 145 cm. The average height of the class is:

(a) 148 cm (b) 147.5 cm (c) 146 cm (d) 147 cm

Ans: (d) $(12 \times 150 + 18 \times 145)/30 = 4410/30 = 147$ cm.

14. A cone has base radius 5 cm and slant height 13 cm. Its height is:

(a) 18 cm (b) 8 cm (c) 12 cm (d) √194 cm

Ans: (c) $h^2 = l^2 - r^2 = 169 - 25 = 144 \Rightarrow h = 12$ cm.

15. If the last step (writing down the final carry) of the addition algorithm is left out, the result for 85 + 47 would be:

(a) 132 (b) 42 (c) 32 (d) 122

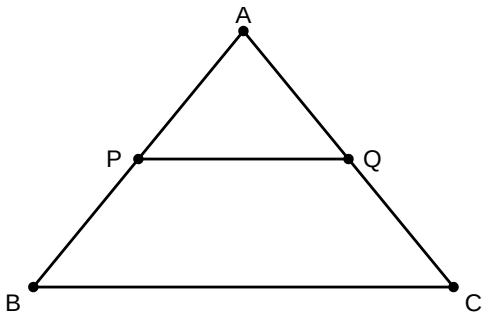
Ans: (c) 5 + 7 = 12 → write 2, carry 1; 8 + 4 + 1 = 13 → write 3, carry 1 (not written) ⇒ 32 instead of 132.

16. In a parallelogram ABCD, ∠A : ∠B = 4 : 5. Then ∠A is:

(a) 90° (b) 72° (c) 80° (d) 100°

Ans: (c) ∠A + ∠B = 180° ⇒ ∠A = 4/9 × 180° = 80°.

17. In △ABC, P is the midpoint of AB and PQ ∥ BC meets AC at Q. If AC = 14 cm, then CQ is:



(a) 10 cm (b) 14 cm (c) 3.5 cm (d) 7 cm

Ans: (d) By the converse of the Midpoint Theorem Q is the midpoint of AC: CQ = 7 cm.

18. Which term of the AP 7, 11, 15, … is 63?

(a) 15th (b) 14th (c) 16th (d) 13th

Ans: (a) $7 + (n - 1) \times 4 = 63 \Rightarrow n - 1 = 14 \Rightarrow n = 15$.

19. **Assertion (A):** The point (−3, 0) lies on the y-axis.
Reason (R): A point on the y-axis has its abscissa equal to zero.

Ans: (d) (−3, 0) has ordinate 0, so it lies on the x-axis; A is false. R is true.

20. **Assertion (A):** The simple average of two group averages equals the combined average only when the two groups are of equal size.
Reason (R): The combined average is the total of all the values divided by the total number of values.

Ans: (a) Both true: R gives the combined average formula $(an + bm)/(n + m)$, which reduces to $(a + b)/2$ only when $n = m$.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A library charges a fine of ₹10 for the first 3 days a book is overdue and ₹4 for every extra day. Find the fine for a book that is 10 days overdue.

Extra days = 10 − 3 = 7 (½)
Extra fine = 7 × 4 = ₹28 (1)
Fine = 10 + 28 = **₹38** (½)

22. Without actual division, state which of $\frac{7}{24}$ and $\frac{7}{32}$ has a terminating decimal expansion. Give reasons.

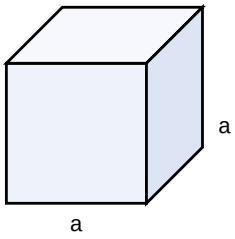
$24 = 2^3 \times 3$ contains the factor 3 ⇒ 7/24 does not terminate (1). $32 = 2^5$ ⇒ **7/32 terminates** (1)

O R

Find two rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.

1/3 = 12/36 and 1/2 = 18/36 (1). Two numbers: **13/36 and 14/36** (any two correct numbers) (1)

23. The total surface area of a cube is 150 cm². Find the length of its edge and its volume.

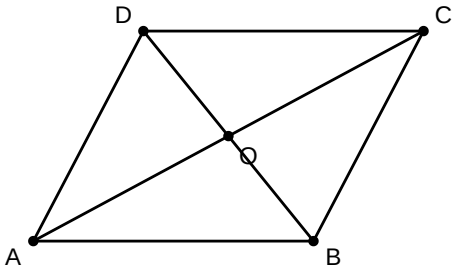


$6a^2 = 150 \Rightarrow a^2 = 25 \Rightarrow$ **a = 5 cm** (1)
Volume = a^3 = **125 cm³** (1)

24. Write the converse of ‘If a number is a multiple of 10, then it is a multiple of 5.’ Is the converse true? Justify.

Converse: **If a number is a multiple of 5, then it is a multiple of 10.** (1)
False: 15 is a multiple of 5 but not of 10. (1)

25. In the parallelogram ABCD, the diagonals meet at O. If OA = (3x − 2) cm and OC = (x + 6) cm, find x and the length of AC.



Diagonals of a parallelogram bisect each other ⇒ OA = OC: 3x − 2 = x + 6 ⇒ **x = 4** (1)
OA = 10 cm ⇒ **AC = 20 cm** (1)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that 0.2777… (7 repeating) = $\frac{5}{18}$.

x = 0.2777…; 10x = 2.777…, 100x = 27.777… (1)
100x − 10x: 90x = 25 (1)
x = 25/90 = **5/18** (1)

27. In a survey of 60 students, the pets they keep were recorded:

Pet	Dog	Cat	Fish	None
Students	20	15	10	15

A student is chosen at random. Find the probability that the student (i) keeps a dog, (ii) keeps some pet, (iii) keeps a cat or a fish.

Total = 60. (i) 20/60 = **1/3** (1) (ii) 45/60 = **3/4** (1) (iii) (15 + 10)/60 = **5/12** (1)

OR

A die was thrown 100 times and the outcomes recorded:

Outcome	1	2	3	4	5	6
Frequency	14	18	16	20	17	15

Find the experimental probability of getting (i) an even number, (ii) a number greater than 4, (iii) a number less than 3.

Total = 100. (i) (18 + 20 + 15)/100 = **53/100** (1) (ii) (17 + 15)/100 = **8/25** (1) (iii) (14 + 18)/100 = **8/25** (1)

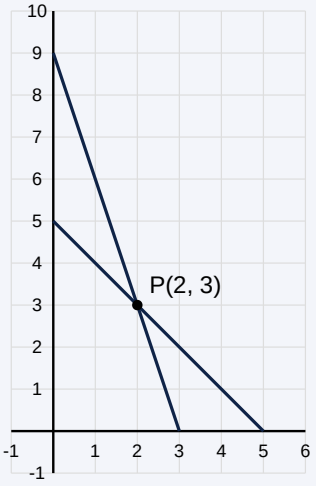
28. Draw the graphs of the equations 3x + y = 9 and x + y = 5 on the same grid and use them to find the solution of the pair.

Table for 3x + y = 9:

x	0	3
y	9	0

Table for x + y = 5:

x	0	5
y	5	0



(1) Draw both lines (1) The lines meet at **(2, 3)**, so x = 2, y = 3 (1)

29. A white stork flew 40 km per day for 3 days and 55 km per day for 2 days. Find its average daily distance over the 5 days and explain why (40 + 55)/2 would be wrong.

Average = (3 × 40 + 2 × 55)/(3 + 2) (1) = 230/5 = **46 km per day** (1)
(40 + 55)/2 = 47.5 treats both speeds as equally weighted although they last for different numbers of days. (1)

30. In a circle with centre O, ∠AOB = 100°. Find ∠ACB, where C is a point on the major arc, and ∠ADB, where D is a point on the minor arc.

∠ACB = ½ ∠AOB = **50°** (1½)
ACBD is a cyclic quadrilateral: ∠ADB = 180° − 50° = **130°** (1½)

OR

Using Heron's formula, find the area of a triangle with sides 25 cm, 52 cm and 63 cm.

s = (25 + 52 + 63)/2 = 70 cm (1)
Area = √[70 × 45 × 18 × 7] (1) = √396900 = **630 cm²** (1)

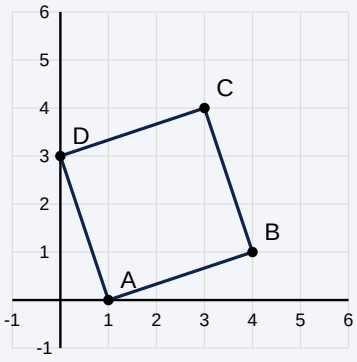
31. Prove that the diagonals of a parallelogram bisect each other.

Let the diagonals of parallelogram ABCD meet at E. (½)
In △AED and △CEB: AD = CB (opposite sides), ∠EAD = ∠ECB and ∠EDA = ∠EBC (alternate angles, AD ∥ BC) ⇒ △AED ≅ △CEB (ASA) (1½)
Hence EA = EC and ED = EB. **The diagonals bisect each other.** (1)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. The corners of a square flower bed are marked on a grid as A(1, 0), B(4, 1), C(3, 4) and D(0, 3).
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths of the four sides. (1½ marks)
(c) Find the lengths of the diagonals AC and BD and justify that ABCD is a square. (1½ marks)



(a) Plotting 4 points (1½), joining (½): (b) AB = √(3² + 1²) = √10; BC = √(1² + 3²) = √10; CD = √(3² + 1²) = √10; DA = √(1² + 3²) = √10 (1½)
(c) AC = √(2² + 4²) = √20; BD = √(4² + 2²) = √20 (1). All sides equal and diagonals equal ⇒ ABCD is a **square** (½).

33. The seats in a hall are arranged in 15 rows. The first row has 18 seats and every next row has 2 seats more than the row before. Find the number of seats in the last row and the total number of seats. (2 + 3)

a = 18, d = 2, n = 15 (1)
Last row = 18 + 14 × 2 = **46 seats** (1)
Total = 15/2 × (18 + 46) (1½) = 15 × 32 = **480 seats** (1½)

O R

A company tested 500 batteries and recorded their lifetimes:

Life (hours)	0 – 100	100 – 200	200 – 300	300 – 400
Number of batteries	30	90	210	170

A battery is chosen at random. Find the probability that it lasts (i) at least 200 hours (1), (ii) less than 100 hours (1), (iii) between 100 and 300 hours (2); (iv) verify that the four class probabilities add up to 1 (1).

Total = 500. (i) (210 + 170)/500 = **19/25** (1) (ii) 30/500 = **3/50** (1) (iii) (90 + 210)/500 = **3/5** (2) (iv) (30 + 90 + 210 + 170)/500 = 1 (1)

34. Prove the converse of the Midpoint Theorem: the line drawn through the midpoint of one side of a triangle, parallel to another side, bisects the third side.

In △ABC, P is the midpoint of AB and the line through P parallel to BC meets AC at Q. To prove: Q is the midpoint of AC. (1)
Let M be the midpoint of AC and join PM. (1)
By the Midpoint Theorem PM ∥ BC (1½)
Through P there is only one line parallel to BC, so PQ and PM coincide ⇒ Q = M (1½)
Hence **AQ = QC**. (½)

O R

A chemist has 200 mL of a 15% acid solution. How many millilitres of a 40% acid solution must be added to obtain a 20% solution?

Let x mL of the 40% solution be added. Acid = 200 × 0.15 + 0.40x = 30 + 0.4x (1½)
Concentration: (30 + 0.4x)/(200 + x) = 0.20 (1½)
30 + 0.4x = 40 + 0.2x (1) ⇒ 0.2x = 10 ⇒ **x = 50 mL** (1)

35. The product of two consecutive positive odd integers is 143. Form a quadratic expression in the smaller integer x, factorise it and find the integers.

The integers are x and x + 2: x(x + 2) = 143 ⇒ x² + 2x − 143 = 0 (1½)
(x + 13)(x − 11) = 0 (1½)
x = 11 (x = −13 rejected) (1) ⇒ the integers are **11 and 13** (1)

OR

A rectangle is formed from 3 square tiles of side x units, 10 strips of size x by 1 unit and 8 unit squares. Write the total area as a polynomial, factorise it and give the dimensions of the rectangle.

> Area = $3x^2 + 10x + 8$ (1)
> $3x^2 + 6x + 4x + 8$ (1½) = $3x(x + 2) + 4(x + 2) = (3x + 4)(x + 2)$ (1½)
> Dimensions: **(3x + 4) units by (x + 2) units** (1)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A shopkeeper stacks tins in a display. The bottom row has 20 tins and every row above it has 2 tins fewer than the row below. The number of tins in the n-th row from the bottom is $t_n = 22 - 2n$.

(i) How many tins are in the first row?

> $t_1 = 22 - 2 =$ **20**

(ii) How many tins are in the 10th row?

> $t_{10} = 22 - 20 =$ **2**

(iii) Find the total number of tins in the display if it has 10 rows.

> Sum = 10/2 × (20 + 2) (1) = **110 tins** (1)

OR

(iii) Which row has exactly 8 tins?

> $22 - 2n = 8$ (1) ⇒ n = 7, i.e. the **7th row** (1)

37. A triangular park has sides 15 m, 41 m and 52 m.

(i) Find the semi-perimeter of the park.

> s = (15 + 41 + 52)/2 = **54 m**

(ii) Find the area of the park using Heron's formula.

> Area = √[54 × 39 × 13 × 2] = √54756 = **234 m²**

(iii) Find the length of the perpendicular from the opposite vertex to the longest side.

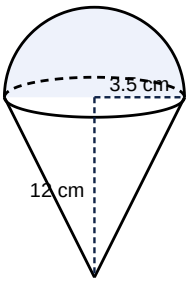
> ½ × 52 × h = 234 (1) ⇒ h = **9 m** (1)

OR

(iii) Grass is laid on the whole park at ₹15 per m². Find the total cost.

> Cost = 234 × 15 (1) = **₹3510** (1)

38. An ice-cream seller serves ice cream in a cone of base radius 3.5 cm and height 12 cm, with a hemispherical scoop of the same radius on top (π = 22/7).



(i) Find the volume of ice cream inside the cone.

> $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 22/7 \times 12.25 \times 12 =$ **154 cm³**

(ii) Find the curved surface area of the hemispherical top.

> $CSA = 2\pi r^2 = 2 \times 22/7 \times 12.25 =$ **77 cm²**

(iii) Find the slant height of the cone and its curved surface area.

> $l = \sqrt{(12^2 + 3.5^2)} = \sqrt{156.25} =$ **12.5 cm** (1); $CSA = \pi rl = 22/7 \times 3.5 \times 12.5 =$ **137.5 cm²** (1)

O R

(iii) Find the total outer surface area of the cone and the hemisphere together (excluding the base).

> Cone CSA = 137.5 cm² (1); total = 137.5 + 77 = **214.5 cm²** (1)